

Ryan Cho

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior software engineer with 6+ years building production Python services and full-stack web applications across finance and SaaS. Designs FastAPI backends, event-driven pipelines, and Vue/TypeScript interfaces backed by PostgreSQL and Snowflake. Ships containerized workloads with Docker, Helm, and Kubernetes, and partners with data scientists and product teams to deliver reliable platforms at scale.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led design of **FastAPI microservices** powering internal portfolio analytics, improving API responsiveness through async query patterns and connection pooling.
- Built **full-stack Vue and TypeScript** dashboards for analyst workflows, standardizing reusable component libraries across product surfaces.
- Authored **ETL pipelines** orchestrating nightly ingestion from PostgreSQL and Snowflake sources, tightening data freshness windows for downstream reporting.
- Containerized services with **Docker and Helm**, deploying to **Kubernetes** clusters across staging and production with zero-downtime rollout strategies.
- Partnered with **data scientists** to productionize model scoring endpoints, establishing contract tests and observability hooks that reduced integration defects.
- Implemented **event-driven integrations** using message queues for settlement notifications, decoupling downstream consumers and improving fault isolation.
- Drove **SQLAlchemy and PostgreSQL** schema evolution for high-volume ledger tables, introducing indexing strategies that sustained predictable read performance under peak load.
- Mentored engineers on **Python service patterns**, code review standards, and deployment workflows, accelerating time-to-first production contribution.
- Collaborated with product managers to translate **stakeholder requirements** into scoped releases, delivering roadmap commitments with predictable velocity.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Python APIs** supporting trading desk workflows, integrating relational stores and downstream risk systems across regional teams.
- Maintained production **SQL and PostgreSQL** reporting pipelines with automated reconciliation and alerting for daily operational datasets.
- Built internal **TypeScript web tools** for operations teams, replacing spreadsheet-driven processes and reducing manual data handling.
- Designed **asynchronous job handlers** for batch settlement tasks, improving throughput during market-close peaks without expanding infrastructure.
- Supported **Docker-based deployments** for application releases, partnering with platform engineering to align service configs across environments.
- Contributed to **ETL workflows** feeding regulatory reporting, documenting data lineage and transformation logic for audit readiness.
- Participated in cross-team architecture reviews for **distributed service boundaries**, advocating maintainable interfaces and versioned API contracts.

- Resolved critical production incidents through structured root-cause analysis, restoring services within tight SLAs during high-volume trading windows.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Implemented **Python REST endpoints** for document processing workflows, enabling reliable uploads and metadata extraction for partner integrations.
- Extended **React front-end modules** for admin dashboards, improving form validation and error feedback across review flows.
- Wrote **SQL migrations and queries** against PostgreSQL, supporting new reporting fields requested by operations stakeholders.
- Containerized development environments with **Docker**, standardizing local setups and shortening onboarding for incoming team members.
- Participated in agile ceremonies and paired with senior engineers on **code reviews**, delivering sprint features with consistent test coverage.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Delivered **responsive web pages** for foundation campaigns using HTML, CSS, and JavaScript, supporting launch timelines for public initiatives.
- Integrated **third-party donation APIs** with lightweight backend hooks, ensuring form submissions persisted reliably to operational databases.
- Optimized **page load performance** through asset minification and lazy loading, improving mobile performance across key landing pages.
- Coordinated content updates with nonprofit stakeholders, translating design mockups into **accessible markup** aligned with WCAG-oriented guidelines.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: TypeScript, Vue, React, JavaScript, HTML5, CSS3, Responsive UI

Backend: Python, FastAPI, REST APIs, SQLAlchemy, Event-Driven Services, ETL Pipelines

Databases: PostgreSQL, Snowflake, SQL, Relational Schema Design

Cloud & DevOps: Docker, Kubernetes, Helm, Terraform, CI/CD, Container Orchestration

Monitoring & Tools: OpenTelemetry, Prometheus, Git, pytest, Argo Workflows

Workflow & PM: Agile/Scrum, Cross-Functional Collaboration, Technical Mentorship, Stakeholder Communication